

2016 H2 Biology Prelim Exam: Paper 3

ANSWER SCHEME

Question 1

A research study conducted in 1979 successfully cloned the human growth hormone (*hGH*) gene using *E.coli* cells. Fig. 1.1 shows the experimental procedure.

- (a) With reference to Fig. 1.1,
(i) identify the enzymes that will need to be added in steps 2 and 5, and state their roles; [3]

Step 2: (ANY 2)

DNA polymerase – to form double stranded cDNA via complementary

base pairing, with cDNA as template

OR

RNase – to digest/remove mRNA

OR

Reverse transcriptase – to form a single stranded cDNA from the mRNA template

Step 5: DNA ligase – catalyzes the formation of phosphodiester bonds between adjacent nucleotides of cut cDNA and plasmid

- (ii) explain the preference of using a restriction endonuclease that produces sticky ends over one that produces blunt ends in step 4. [2]
1. Sticky ends will allow cut DNA fragments to anneal spontaneously
 2. by forming hydrogen bonds with complementary sticky ends of DNA fragments cut up by the same enzyme, unlike
 3. blunt ends which make annealing more difficult/ requires an additional step
 4. of ligating sticky ends/ linker DNA to the restriction fragments for hydrogen bonds to form.

In a separate experiment using the same procedure as Fig. 1.1, a pBR322 plasmid was used and the gene was inserted into the tryptophan (*trp*) operon between the structural genes. This allows for high levels of the hormone to be produced.

- (b) Suggest how the expression of the *hGH* gene can be controlled. [2]

1. Control of production of the hormone/ expression of gene through manipulating the growth medium – exclude tryptophan.
2. The trp operon can be switched on/expressed, thus producing the hormone/protein.

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cDNA libraries can also be formed using isolated mRNA.

(c) Outline the uses of cDNA libraries. [3]

1. To deduce the amino acid sequence of the protein from the DNA sequence.
2. To produce a desired protein in bulk (e.g. human growth hormone, insulin, etc.) by expressing the cloned gene in a bacterial or yeast cell.
3. To trace changes in the patterns of gene expression - By making cDNA from cells of the same type at different times in the life of an organism

In rare situations, *hGH* gene may be obtained from bacterial cells in a genomic DNA library.

(d) Describe the steps that can be taken to identify the bacterial cells with the *hGH* gene. [3]

1. Bacterial colonies on the original plate are replicated/ transferred onto a nitrocellulose filter.
2. The filter with the bacterial colonies is treated with NaOH to lyse the bacterial cells and denature the DNA to form single stranded DNA.
3. After washing with chloroform and saline, it is baked for a few minutes.
4. A specific radioactive gene probe, which has a sequence complementary to part of the gene of interest, is added. Hence it would hybridize/ bind with DNA which contain the gene of interest.

Unbound probes are washed away.

5. The position of colonies with the bound probes can be located by autoradiography.
6. The autograph is compared against the master plate and the bacteria colonies with the gene of interest are identified.

[Total: 13]

Question 2 [13 marks]

In gene therapy, normal functional alleles are introduced into a person who has defective genes which do not produce an important protein. An experiment was done to compare the effectiveness of the following treatments.

1. The protein was injected directly into the tissues.
2. Harmless viruses carrying normal functional allele coding for the substance were injected.
3. The normal functional alleles were put into a protein capsule which was inserted into the tissues.

Fig. 2.1 shows results of the experiments.

Fig. 2.1

- (a) Using the information in Fig. 2.1, suggest **one** advantage and **one** disadvantage of the protein capsule treatment. [2]

The advantages:

1. It does not result any harmful side effects / It is less likely to harm the person, longer relief from symptoms
OR
2. It remains effective for a longer period of time / Less frequent treatment needed

Disadvantages

1. It takes a longer time to take effect / for the tissues to produce the protein.
2. It has a lower effective concentration range.

Fig. 2.2

- (b) With reference to Fig. 2.2, **explain** the changes in white blood cells. [3]

- 1a. 0 to 2 months after injection of stem cells, the number of functioning white blood cells remains constant 100 per mm³ of blood
- 1b. because it takes time for the production of new white blood cells.
- 2a. 2 to 10 months after injection of stem cells injection, the number of functioning white blood cells per mm³ increases from 100 per mm³ of blood to 5800 per mm³ of blood
- 2b. because the modified stem cells with the normal functional allele has differentiated to form any functional white blood cells.
- 3a. 10 to 12 months after stem cells injection, the number of functioning white blood cells remains constant 5800 per mm³ of blood
- 3b. because the body has produced the normal amount of white blood cells.

(c) Explain why stem cells were used in the treatment. [3]

1. They can divide and self-renew
2. So there will be many modified stem cells in the body for a long time
3. They can differentiate
4. to form any type of blood cell with the normal functional allele / functional white blood cells
To restore ability to fight infection;
5. The blood stem cells were the child's own cells;
6. So will not trigger immune response / cause rejection

(d) Explain why it is sufficient to insert one copy of the normal *ADA* allele into each blood stem cell to treat SCID. [2]

1. SCID is an autosomal recessive disorder / the mutant allele is recessive. [1]
2. addition of only one copy of normal functional allele by gene therapy will lead to production of normal functional protein.
3. This will mask the effect of recessive allele.

(e) Suggest why the normal functional allele in gene therapy is likely obtained from a cDNA library rather than the genomic DNA library. [3]

1. cDNA library contains genes made from mRNA through the process of reverse transcription
WHILE
2. Genomic DNA library contains fragments of DNA derived from entire genome of an organism
3. cDNA library contain intact genes
4. cDNA library only contain genes which are expressed
5. So it is smaller and easier to find the desired cDNA gene
6. Gene probe is required to find the desired gene in the genomic DNA library.

Also accept:

7. cDNA library contain genes without introns
8. The gene is a shorter sequence hence higher probability of transfection (successful gene delivery)
9. Also gene product can be formed in a faster time – no need for RNA splicing

Question 3

Golden Rice is a genetically modified crop that is developed by scientists to express two genes that are involved in the biosynthesis of β -carotene.

(a) Explain why there was a need to develop Golden Rice. [2]

1. Current problem: Many people, especially in developing countries, suffer from vitamin A deficiency
2. Since rice is often a staple in countries with high incidence of vitamin A deficiency
3. Golden rice provides people with β -carotene which can form vitamin A / provide vitamin A in their diet / was developed with the goal of helping millions of people who suffer from vitamin A deficiency / OWTTE
4. which can prevent blindness and death.

Golden rice was genetically modified by inserting 3 different DNA sequences into plasmids. The 3 sequences were:

1. Psy cDNA genes obtained from daffodils
2. Glu, a rice endosperm-specific glutelin promoter
3. crt I genes from *E. uredovora*

The recombinant plasmids were then inserted into different cultures of rice protoplasts.

(b) Suggest why the Glu promoter was inserted into the plasmids. [1]

Any one:

1. To ensure that the inserted genes will be expressed in the endosperm of the rice.
2. A suitable promoter may not be present in the endosperm of the rice.

(c) Outline how genetically engineered Golden Rice protoplasts can develop into full-grown crops. [5]

(Stage 1: Establishment)

1. Place the protoplasts in a sterile nutrient medium containing the following:
2. a carbon and energy source, organic and inorganic nutrients, and plant growth regulators.
3. The plant growth regulators will stimulate mitosis/ cell division and
4. result in the formation of a callus from the cells placed in the medium.

(Stage 2: Multiplication)

5. If the explant is successfully established, several microshoots are produced in a few weeks.
6. The established and stabilized culture is transferred to a nutrient medium with an increased cytokinin and decreased auxin concentration to encourage shoot initiation.
7. After sufficient microshoots have grown, the cytokinin concentration of the culture is changed to encourage shoot elongation.

(Stage 3: Rooting)

8. After the microshoots have reached an appropriate length, they are transferred to a root inducing medium where the increased auxin and reduced cytokinin concentration encourages root formation.

(Stage 4: Acclimatization)

9. Shift the plantlet from a heterotrophic (sugar-requiring) to an autotrophic (free-living) condition,
10. and acclimatize the plantlet to the outdoor environment / in the farmland/soil/paddy fields.

In a study to determine the most effective source of vitamin A for children, 3 groups of children in China were given different sources of β -carotene. The 3 sources were

1. Golden Rice
2. β -carotene capsules
3. Spinach (a dark-green leafy vegetable that is rich in β -carotene)

Before the study, baseline levels of vitamin A in the children were measured. Children in each group were given one source of β -carotene and the vitamin A levels were measured again after 1 day.

To better ascertain the effectiveness of different sources of β -carotene, scientists have also calculated the efficiency of conversion of β -carotene to vitamin A.

Scientists have concluded that Golden Rice is the most effective source of β -carotene.

Table 3.1

Source of β -carotene	β -carotene capsule	Golden Rice	Spinach
Amount of β -carotene in one dose (mg)	0.5	0.6	1.4
Mean baseline level of vitamin A ($\mu\text{g/dL}$)	23.6	25.6	24.2
Mean level of vitamin A 1 day after dose was administered ($\mu\text{g/dL}$)	24.1	26.2	24.5
Conversion efficiency of β -carotene to vitamin A (%)	11.1	43.4	4.8

(d) With reference to Table 3.1, explain how the results of the study support the scientists' conclusion. [3]

1. [evaluation of trends and QF] Children who ate Golden Rice showed the greatest increase in vitamin A levels, from 25.6 $\mu\text{g/dL}$ to 26.2 $\mu\text{g/dL}$
2. as compared to children who ate β -carotene capsule, where vitamin A levels increased from 23.6 $\mu\text{g/dL}$ to 24.1 $\mu\text{g/dL}$, or spinach, where vitamin A levels increased from 24.2 $\mu\text{g/dL}$ to 24.5 $\mu\text{g/dL}$.

3. This is also shown in the greatest conversion efficiency of β -carotene to vitamin A for Golden Rice, at 43.4%, as compared to 11.1% for β -carotene capsule and 4.8% of spinach. [1]
4. These data suggests the Golden Rice is the most effective source of β -carotene, as most β -carotene is converted to vitamin A. [1]

(e) Discuss the social and ethical implications of the cultivation and consumption of Golden Rice. [3]

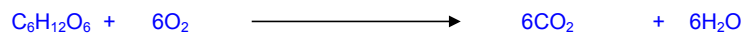
1. Effectiveness of vitamin A absorption and function in the body
 - Vitamin A deficiency is usually caused by a lack of a balanced diet.
 - Because carotenes are hydrophobic, they require a certain level of fat in the diet and in the body to function well.
 - In poor, developing countries where people are malnourished, consumption of Golden Rice may be ineffective as they do not have the necessary fat content in their diet.
2. Possible threats to the environment
 - It is possible that the cross-pollination of golden rice with the wildtype can potentially introduce genes from golden rice to the wild rice gene pool.
 - There has been no extensive and conclusive evidence to ascertain if the growing and harvesting of Golden Rice would have an impact on the environment.
3. Alternative sources of vitamin A
 - Many have argued that there are other natural, cheaper and risk-free sources of vitamin A found in green leafy vegetables and unpolished rice. Hence, the growing and consumption of Golden Rice is an unnecessary expense and a waste of money and resources.

Question 4

INTRODUCTION

A. Rationale [max 1 mark]

1. Mealworms undergo respiration in the presence of the carbohydrates/ glucose according to the following equation.



2. The rate of respiration is affected by temperature, as enzymes in glycolysis, link reaction, Krebs cycle and oxidative phosphorylation are affected by temperatures.
3. CO₂ is a gas given off during oxidative decarboxylation reactions, and hence its rate of production is a reflection of rate of respiration of the mealworms.
4. Therefore, the higher the temperature (below optimum temperature), the more carbon dioxide is given off per unit time, and hence the higher the rate of respiration.

METHODOLOGY

B. Variables & Controlled Variables

[1/2 mark each]

IV: Independent variable – temperature (°C)

DV: Dependent variable – volume of carbon dioxide (ppm)

[any 2; ½ mark each]

CV: Controlled variables:

1. Number and weight of mealworm larvae and beetles used
 - It affects the volume of CO₂ formed.
 - Keep the number and weight constant using the electronic balance.
2. Duration of reaction
 - It affects the amount of enzyme-substrate complexes that can be formed, and hence the volume of CO₂ formed.
 - A digital stopwatch is used to ensure duration of reaction is kept constant
3. Concentration of oxygen
 - It affects the volume of CO₂ formed.
 - Kept constant as the concentration of oxygen in the air is approximately 20%.

C: DETAILED PROCEDURE [4 marks max]

1. Set up the thermostatically-controlled incubator to 10°C.
2. Weigh 2-3 mealworm larvae using the weighing balance and note its weight.

3. Using forceps, transfer the larva into the glass jar and attach the carbon dioxide sensor.
4. Place the glass jar with sensor into the incubator.
5. Start the stopwatch and allow for 1-2 minutes for equilibration.
6. After 2 minutes, read the carbon dioxide volume off the sensor and recording this reading as the initial carbon dioxide volume.
7. Re-start the stopwatch and time for 5 minutes.
8. After 5 minutes, record the final carbon dioxide volume.
9. Repeat steps 1 to 8 to obtain a total of three readings, using larvae of similar weights each time.
10. Repeat steps 1 to 9 for another 6 temperatures – 20°C, 30°C, 40°C, 50°C, 60°C and 70°C.
11. Repeat steps 1 to 10 for 2-3 mealworm beetles, using similar weights for each repeat.
12. To show that the carbon dioxide given off is due to the respiring mealworm larva or beetle, a control is set up. Steps 1 to 11 are performed without the mealworm larva or beetle. The control is subjected to the same environmental factors as that for the experiment.
13. To ensure reproducibility of data, repeat the entire experiment twice (steps 1 to 12) using mealworm larvae and beetles of a different batch.

D: DATA MANIPULATION AND EXPECTED RESULTS

For mealworm larvae,

For mealworm beetle,

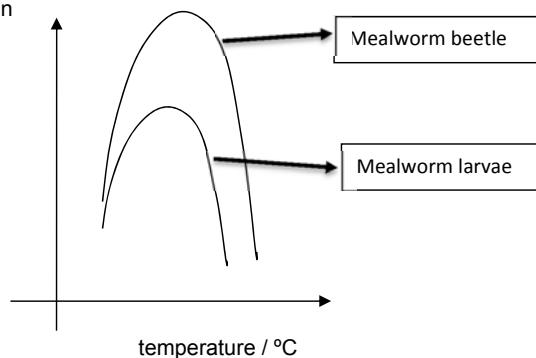
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70										

2. Graphs [1 mark]

Graph of average volume of CO₂ given off in 5 minutes / ppm against temperature / °C

average vol of CO₂ given

off in 5 minutes / ppm



3. [Describe and Explain the expected trends and/or results] [2 marks]

- [describe similarity]** For both mealworm larvae and beetle, as temperature increases, the average volume of CO₂ given off in 5 minutes increases till optimum temperature (at about 35-40 °C) then decreases at temperatures beyond optimum temperature.
- [explain similarity]** Increase in temperature increases the rate of effective collisions between enzymes and substrates of respiration. More enzyme-substrate complexes form per unit time. Hence rate of respiration increases and more CO₂ is given off per unit time.
- [describe difference]** The average volume of CO₂ given off in 5 minutes for beetle is more than the larvae.
- [explain difference]** The beetle is the adult mealworm, which carries out more complex metabolic reactions which require more energy/ATP. Therefore, rate of respiration is higher to match ATP demand, as compared to the larvae. / OWTTE

SAFETY PRECAUTIONS [any 2; ½ mark each]

- Handle the glass jar with care as it is prone to breakage when dropped from a height. Dispose of broken glass shards in the sharps bin.
- Handle live specimen with gloves, forceps carefully and ensure they are always kept in a closed container.
- Handle the thermostatically-controlled water bath and hot glass jar with care at the higher temperatures. Use a rag to handle the apparatus.

Question 5

- (a) Describe how gel electrophoresis can be used to provide evidence of molecular homology. [8]
- Closely related species have more similar nucleotide sequences between their DNA than distantly related species. [1]
OR
The more recently two species branched from a common ancestor, the more similar their DNA and amino acid sequences would be.

B: Also mutations in non-coding regions or silent mutations are usually passed on to subsequent generations since these mutations do not affect the survival of the organism.

B: Species with more of these mutations are likely to be more closely related.
 - Obtain DNA samples from the cells of different species
 - Use same restriction enzymes to digest DNA.
 - The collection of restriction fragments from each species will result in a unique band pattern upon gel electrophoresis [1]
 - Restriction fragments are separated using gel electrophoresis
 - Load DNA samples into separate wells located at negative electrode
 - DNA is negatively charged due to phosphate groups and under influence of direct current, the fragments move to positive terminal. [1]
 - Agarose gel forms cross-link matrix which offer resistance to fragments
 - Longer fragments encounter more resistance and move slower / Shorter fragments encounter less resistance and move faster and further.
 - Fragments are separated according to size.
 - The banding patterns are compared.
 - Species which are more closely related will have more bands at similar positions. [1]
- (b) Explain how RFLP analysis facilitates the processes of genomic mapping in terms of linkage mapping [6]
- A genetic / linkage map is an ordered list of genes / genetic markers along a chromosome that
 - includes the order and relative distances between those genes / genetic markers.
 - The relative distances is based on recombinant frequencies.
 - Recombinant frequency is the frequency at which crossing over occurs between two loci on a pair of homologous chromosome.
The frequency of recombination is a measure of genetic distance between two loci on a DNA molecule.

5. The lower the recombination frequency, the closer the two loci are.
6. To calculate the relative genetic distance between two loci on the same chromosome, RFLPs are used to measure recombination rates (crossing over). This is because RFLPs
7. RFLP act as genetic markers and provide increased number of markers besides the use of gene as markers.
8. RFLPs are usually found in non-coding regions. Using RFLPs as markers allows mapping of non-coding regions
9. Also allows mapping of regulatory sequences
Length of restriction fragments formed allows estimation of number of nucleotides between markers / genes.
10. The banding patterns are analysed and counted among the offspring.
11. With RFLP analysis, the relative genetic distance between two RFLP loci, is measured by noting the position of their bands on a gel.
12. A genetic / linkage map with the distance between RFLP loci measured in centiMorgans (cM) can then be produced.

(c) Discuss the ethical concerns that have arisen about the human genome project. [6]

Any combinations to obtain max 6 marks

Issues regarding evaluation and regulation of genetic tests [2]

1. There are concerns that some genetic tests for the detection of diseases may not be very accurate / reliable.
2. There are concerns that standards and quality-control measures may not be properly implemented during genetic testing procedures.
3. All these may lead to false positive results / false negative results.
4. There are also concerns that genetic tests are marketed to consumers directly via the internet. As a result, customers receive clinically significant test results without appropriate counselling from a healthcare professional, and are left to interpret results on their own.

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Issues concerning support given by clinics to help patients make informed decisions and give informed consent [2]

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5. Proper counselling by certified medical professionals should be given to patients about the limitations of genetic tests and risk gene therapy.
6. Couples about the use of genetic information in reproductive decision making and regarding reproductive rights.
7. Therefore, doctors and other healthcare professionals should be trained in the dissemination of genetic information and patient counselling in order to raise the standards of patient care.
- 4.8. So that patients can make informed decisions about their own healthcare and how best to proceed.

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Issues regarding the use of genetic information in reproductive decision making and regarding reproductive rights [1]

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4.9. There are concerns that genetic testing of an unborn fetus might lead to parents aborting the pregnancy if the fetus is tested positive for certain diseases e.g. Down syndrome.

10. This issue concerns the moral status of the embryo and some may consider abortion as tantamount to killing a human life.

Issues regarding the identification of complex diseases and diseases with no cure [2]

11. There is concern that genetic testing for conditions linked to multiple genes and gene-environment interactions, e.g. heart disease, which currently have no treatments nor cure

12. This may lead to negative impact on one's psychological and emotional well-being.

This may lead to genetic discrimination.

Issues regarding genetic discrimination / stigmatization [2]

13. There is concern that individuals tested positive for genetic diseases become socially marginalized individuals that are deemed to be "genetically inferior". They may be discriminated against by insurance companies or by employers.

4. 15. Insurance companies may use the genetic information to deny or limit insurance coverage for individuals who are genetically pre-disposed to diseases that they may develop at a later stage.

16. Employers may use the genetic information to hire or retain only individuals who are not genetically pre-disposed to future disease onset as they are perceived to be more productive.

Issue regarding ownership of genetic information [2]

17. Questions have been raised as to whether genetic information (e.g. specific gene sequences) is owned by individuals or by biotechnology companies.

18. Patents issued for new genetic discoveries encourage innovation by private biotech companies and allows for proper sharing of inventions and discoveries.

19. However other researchers are not given access to the patented genetic information (e.g. specific gene sequences) and this limits the progress of biomedical research.

20. It also prevents genetic discoveries from translating into healthcare benefits.

Issue regarding privacy and confidentiality of genetic information [2]

21. Questions have been raised as to who should have access to genetic information.

22. Genetic information of individuals should be protected so that they are not harmed as a result of improper disclosure and misuse of information.

23. There are concerns that the genetic information will be made accessible to insurance companies, employers, lawyers, the military, schools, hospitals and biotechnology companies.

24. There are concerns that confidentiality / non-disclosure could mean that other family members would not be informed that they might also carry the genetic defect.

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